

Hassan Hosny Shawky

Mechatronics Engineering Student || Asyut, Egypt

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Profile

Dedicated third-year Mechatronics Engineering student at Sphinx University with a cumulative GPA of 3.83/4.0 and a Distinction academic record for three consecutive years. Experienced in embedded systems, industrial automation, and real-world project development. Completed intensive training at HA Consulting Group covering Classical Control, PLC programming, Motor Drive, and EPLAN — with 60+ hands-on projects. Passionate about integrating Artificial Intelligence with industrial control systems to advance production line automation and robotics.

Education

Bachelor of Science in Mechatronics Engineering

2022 – 2028

Sphinx University — Asyut, Egypt

3rd Year Student | Cumulative GPA: 3.83 / 4.0 | Academic Standing: Distinction — 3 Consecutive Years

Training

Industrial Automation Diploma | HA Consulting Group

Jun 2025 – Aug 2025

- Completed practical training in Classical Control Systems (theoretical & practical).
- Programmed PLCs across four platforms: Siemens, Schneider, Wecon, and Delta.
- Configured and operated Motor Drive & Inverters (VFD): Yaskawa J1000, LS iC5 (SV008iC5-1), LS M100 (LSLV0022M100-1E0FNS), and SAVCH S800 (S800-2S0.75G).
- Designed electrical schematics using EPLAN Electric P8.
- Completed 60+ hands-on projects covering packaging systems, color & object sorting, and motor speed & direction control.

Projects

Industrial Automated Sorting System | PLC Simulation & Factory I/O

- Developed a simulated production line that automatically sorts three product types based on shape and color using Siemens PLC Simulation (SIMATIC Manager) and Factory I/O.
- Implemented sensor-based product detection with memory storage and value comparison to identify product types and activate the corresponding sorting arm.
- Programmed in Ladder Logic with full control features: Start/Stop, Emergency Stop, Counter Reset, three individual product counters, and Automatic/Manual mode selector.
- Manual mode allows main conveyor operation independently without activating sorting arms.

Tank Level Control & Monitoring System | Siemens PLC, HMI & Factory I/O

- Developed a tank level control and monitoring system using Siemens PLC, HMI, and Factory I/O simulation.

- Implemented analog water level measurement with automatic tank filling and discharge control based on an adjustable setpoint configured through the HMI.
- Designed HMI screens featuring real-time level visualization, alarm monitoring, and trend graphs to track level changes over time.
- Programmed in Ladder Logic; integrated PLC, HMI, and Factory I/O into a complete industrial control and monitoring system.

Star-Delta Motor Control System | Siemens S7-300 PLC & HMI

- Designed and programmed a Star-Delta motor starter control system using Siemens S7-300 PLC and SIMATIC Basic Panel HMI, replacing traditional push buttons with full HMI-based control.
- Programmed control logic in Ladder Logic (LAD) to ensure correct Star/Delta sequencing and prevent forward/reverse direction conflicts.
- Implemented real-time status monitoring via HMI indicator lamps displaying Main Forward, Main Reverse, Star, and Delta states.
- Validated full system logic using RT Simulator prior to real-world implementation.

Size-Based Automated Sorting System | Siemens PLC & Factory I/O

- Developed an automated production line that classifies and sorts three product types (small, large, long) based on dimensions using digital sensor combinations and Siemens PLC (SIMATIC Manager).
- Implemented sensor logic: small products detected by sensor 1 only; large products by sensors 1 & 2; long products by sensors 1 & 3 — each routed to its designated conveyor via pneumatic pushers.
- Programmed in Ladder Logic (LAD) with individual product counters, a total batch counter, and fully automatic conveyor and sorting control.
- Simulated the complete production environment using Factory I/O for logic validation before deployment.

Color-Sorting Production Line

- Designed and built a fully automated production line that detects and sorts objects by color in real time.
- Hardware: Arduino Nano, color sensor, motion sensors, servo motor, DC motor, and LCD display.

Smart City System

- Developed a microcontroller-based smart city model integrating automated control of agricultural irrigation, smart lighting, residential door locks, and traffic management.

Hybrid Solar & Wind Energy Station

- Built a hybrid energy station combining solar and wind sources with intelligent source-switching for maximum daily efficiency.
- Integrated a solar tracking mechanism to follow the sun's position throughout the day for optimal panel output.

Smart Garage & Smart Car

- Smart Garage: Automated entry control system with sensor-based detection and motorized gate operation.
- Smart Car: Microcontroller-driven vehicle with obstacle avoidance and line-following capabilities.

Skills

- PLC Programming: Siemens, Schneider, Wecon, Delta.
- Motor Drive & Inverters (VFD): Yaskawa, LS Electric, SAVCH.
- EPLAN Electric P8 — Electrical Schematic Design.
- Classical Control Systems — Theoretical & Practical.
- Embedded Systems: ESP32, Arduino (Uno, Nano).
- 3D CAD: SolidWorks.
- Electrical components knowledge and soldering.

Additional Information

Languages: Arabic (Native), English (Technical Proficiency).

Interests: Robotics, AI-driven industrial control, production line automation, renewable energy systems.